

Factorization of $\sum_{i=0}^k x^{in}$: Roots of Unity, False Generalization, and Cyclotomic Structure

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§1.1 The original problem and the first successful method

The paper begins with the factorization of

$$x^{2n} + x^n + 1$$

by

$$x^2 + x + 1.$$

Let ω be a primitive cubic root of unity. Then

$$\omega^2 + \omega + 1 = 0, \quad \omega^3 = 1, \quad \omega \neq 1.$$

The polynomial $x^2 + x + 1$ divides $x^{2n} + x^n + 1$ iff both ω and ω^2 are roots. Substituting $x = \omega$ gives

$$\omega^{2n} + \omega^n + 1.$$

This is zero exactly when ω^n is a nontrivial cubic root of unity. Therefore the divisibility holds iff

$$n \not\equiv 0 \pmod{3}.$$

This proof is short because roots of unity convert a divisibility question into an exponent congruence question.

§1.2 The direct factorization attempt

The note then tries to prove the same statement by direct algebra. Since

$$x^{2n} + x^n + 1 = \frac{x^{3n} - 1}{x^n - 1},$$

one can try to factor numerator and denominator using difference of squares or cubes. For example, when $n = 2$,

$$x^4 + x^2 + 1 = (x^2 + x + 1)(x^2 - x + 1).$$

When $n = 4$, one obtains several quadratic-looking factors, including again $x^2 + x + 1$.

But when n changes parity or becomes divisible by 3, the algebraic route splits into cases. The original article carefully lists cases such as $n = 3m \pm 1$ and tracks which geometric-series quotients appear. The lesson is that direct factorization can prove the theorem, but it obscures the simple reason: the exponent map $z \mapsto z^n$ acts on the three roots of unity.

§1.3 The attempted generalization

The natural next question is whether

$$1 + x^n + x^{2n} + \cdots + x^{kn}$$

is divisible by

$$1 + x + x^2 + \cdots + x^k$$

whenever n is not a multiple of $k+1$. This first guess is too weak. The correct condition is coprimality.

Theorem 1.3.1 (Root-of-unity test for the generalization)

Over a field of characteristic zero,

$$1 + x + x^2 + \cdots + x^k \quad \text{divides} \quad 1 + x^n + x^{2n} + \cdots + x^{kn}$$

if and only if

$$\gcd(n, k+1) = 1.$$

Proof. The roots of

$$1 + x + x^2 + \cdots + x^k = \frac{x^{k+1} - 1}{x - 1}$$

are exactly the nontrivial $(k+1)$ -st roots of unity. Let $\xi \neq 1$ satisfy $\xi^{k+1} = 1$. We need

$$1 + \xi^n + \xi^{2n} + \cdots + \xi^{kn} = 0$$

for every such ξ .

This is a geometric sum. It vanishes exactly when ξ^n is a nontrivial $(k+1)$ -st root of unity. Therefore the map

$$\xi \mapsto \xi^n$$

must send every nontrivial $(k+1)$ -st root to another nontrivial $(k+1)$ -st root. Equivalently, the power map on the cyclic group μ_{k+1} must have trivial kernel. This happens exactly when n is invertible modulo $k+1$, i.e.

$$\gcd(n, k+1) = 1.$$

□

The original false guess $n \not\equiv 0 \pmod{k+1}$ fails because a nonzero residue may still not be invertible. For instance, modulo 6, the residue 2 is nonzero but the map $z \mapsto z^2$ has a nontrivial kernel on sixth roots of unity.

§1.4 The computational experiment

The note then turns to MATLAB. It computes roots of

$$x^{kn} + x^{(k-1)n} + \cdots + x^n + 1 = 0$$

for different values of n and plots them in the complex plane, sometimes using a third coordinate to record n . This is not just decoration. The picture reveals periodicity in the arguments of the roots.

The observed pattern is: the roots lie on the unit circle, their arguments are fractions of 2π , and as n varies the same angular patterns repeat. For $k = 2$, the pattern has two angles per block; for $k = 3$, three angles per block; in general, the $(k + 1)$ -st-root structure controls the blocks.

This experimental section is important because it records a genuine mathematical process: a false conjecture is tested, numerical pictures suggest the right invariant, and then the invariant is expressed exactly by roots of unity.

§1.5 Explicit root formula

The equation

$$1 + x^n + x^{2n} + \cdots + x^{kn} = 0$$

can be rewritten as

$$\frac{x^{n(k+1)} - 1}{x^n - 1} = 0.$$

Thus the roots are those $n(k+1)$ -st roots of unity whose n th power is not 1. Equivalently,

$$x = \exp\left(\frac{2\pi i}{n(k+1)}(r(k+1) + m)\right),$$

where $r = 0, \dots, n-1$ and $m = 1, \dots, k$. This gives exactly kn roots, as expected from the degree.

The original note writes these roots as products of exponentials:

$$e^{2\pi i r/n} e^{2\pi i m/(n(k+1))}.$$

Substituting into the polynomial gives a finite geometric sum whose ratio is a nontrivial $(k+1)$ -st root of unity, hence the sum is zero.

§1.6 Common real factors

The final theorem concerns common real-coefficient factors among these polynomials as n varies. Since complex roots come in conjugate pairs, a set of roots stable under conjugation gives a real polynomial factor. When $d = \gcd(n, k+1) > 1$, some of the angular positions line up across different n , producing common roots and hence common real factors.

This is the cyclotomic viewpoint in disguise. The polynomial

$$x^m - 1$$

factors as

$$x^m - 1 = \prod_{d|m} \Phi_d(x),$$

and the geometric sum

$$1 + x + \cdots + x^k = \frac{x^{k+1} - 1}{x - 1}$$

contains the cyclotomic factors $\Phi_d(x)$ for divisors $d > 1$ of $k+1$. Replacing x by x^n pulls these cyclotomic conditions back along the power map.

§1.7 The organizing idea

The value of this note is the exploration, not only the final condition. It begins with a correct theorem, tries a direct factorization proof, guesses a generalization, finds that the guess is false, uses computation to see the roots, and finally explains the pattern by complex exponentials. That is a miniature version of mathematical research: algebraic manipulation, conjecture, counterexample, visualization, and structural explanation.

§1.8 Cyclotomic summary of the experiment

The computational pictures, the failed generalization, and the final divisibility criterion all express one group-theoretic fact: roots of unity form finite cyclic groups, and power maps on cyclic groups are controlled by greatest common divisors.

Remark 1.8.1 (From roots to factors) — A rational polynomial divisibility statement can be checked on all Galois conjugate roots. For roots of unity, the conjugates of a primitive m -th root are

$$\zeta_m \mapsto \zeta_m^a, \quad a \in (\mathbb{Z}/m\mathbb{Z})^\times.$$

Thus a common factor is not merely one common complex root; it is a conjugation-stable set of common roots.

The cyclotomic factorization

$$x^m - 1 = \prod_{d|m} \Phi_d(x)$$

separates roots by exact order. Geometric sums select all nontrivial roots of a given order divisor, while substitutions $x \mapsto x^n$ pull those root conditions back along the power map. This is the structural explanation of the root plots in the original experiment.

§1.9 Cyclotomic polynomials

The roots of $x^n - 1$ are the n -th roots of unity. The primitive n -th roots are those with exact order n . The cyclotomic polynomial is

$$\Phi_n(x) = \prod_{\substack{1 \leq k \leq n \\ \gcd(k,n)=1}} (x - e^{2\pi i k/n}).$$

It has degree $\varphi(n)$ and integer coefficients.

The factorization is

$$x^n - 1 = \prod_{d|n} \Phi_d(x).$$

This identity is the structural version of many elementary root-of-unity factorization exercises.

§1.10 Galois action on roots of unity

The field $\mathbb{Q}(\zeta_n)$ generated by a primitive root of unity ζ_n is the n -th cyclotomic field. Its Galois group is

$$\text{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \cong (\mathbb{Z}/n\mathbb{Z})^\times.$$

An element $a \in (\mathbb{Z}/n\mathbb{Z})^\times$ acts by

$$\zeta_n \mapsto \zeta_n^a.$$

Thus the same reduced residue system that appears in elementary number theory controls the symmetries of roots of unity.

§1.11 Resultants and common roots

Two polynomials have a common root exactly when their resultant is zero. For example, studying when a cyclotomic polynomial divides another expression can be reframed as asking which roots are common.

The elementary method says: substitute a primitive root ω and check whether the expression vanishes. The algebraic version says: divisibility is detected by vanishing on all Galois conjugate roots, or by a resultant condition.

§1.12 Finite fields and character sums

Roots of unity also appear in finite fields. The multiplicative group $\mathbb{F}_q^\times$ is cyclic of order $q - 1$. Therefore roots of $x^n = 1$ in $\mathbb{F}_q$ are controlled by $\gcd(n, q - 1)$.

Character sums such as

$$\sum_{x \in \mathbb{F}_q} \chi(x) \psi(f(x))$$

are research-level descendants of elementary root-of-unity sums. They connect finite fields, Fourier analysis on finite groups, and algebraic geometry.

§1.13 Returning to the original experiments

The note's factorization experiments with roots on the complex plane are the concrete beginning of cyclotomic theory. The advanced bridge is precise: roots of unity form finite cyclic groups, cyclotomic polynomials package primitive roots, and Galois groups act by exponentiation modulo n .

§1.14 Primitive roots and cyclotomic factor tests

To decide whether a polynomial expression is divisible by $\Phi_n(x)$, it is enough to substitute a primitive n -th root of unity ζ and check whether the expression vanishes. Because Φ_n is the minimal polynomial of ζ over $\mathbb{Q}$, vanishing at one primitive root forces vanishing at all Galois conjugates.

This explains why root-of-unity substitution is not just a clever complex-number trick. It is a minimal-polynomial test.

§1.15 Cyclotomic identities through Möbius inversion

From

$$x^n - 1 = \prod_{d|n} \Phi_d(x)$$

one obtains

$$\Phi_n(x) = \prod_{d|n} (x^d - 1)^{\mu(n/d)}.$$

This is Möbius inversion on the divisor lattice. The alternating exponents in this formula are the same inclusion–exclusion pattern seen in counting and arithmetic functions.

Thus the note’s factorization experiments sit at an intersection of complex roots, integer polynomials, Galois symmetry, and Möbius inversion.